Shimanto Bhowmik

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Education

Rochester Institute of Technology — Rochester, N.Y.
Bachelor of Science in Computer Science. GPA: **3.99/4.0** — *Aug. 2021 – Dec. 2025*

Work Experience

Software Engineer — April 2026 – Present
Amazon Web Services — *Seattle, WA.*

- AWS Security

Software Engineer — Jan. 2026 – April 2026
FinPilot — *Seattle, WA*

- Built a **schema mapping engine** and **differential sync pipeline** for CRM integrations (Bipsync, Altvia), automating onboarding for **3 enterprise clients**, reducing setup time by **85%** and lowering data staleness to $<$ **1 hour**.
- Designed and launched a greenfield **performance tracking module** using **LLMs** to auto-populate return metrics, support manual overrides and link funds across portfolios.

Software Engineering Intern — May 2025 – Aug. 2025
Amazon AGI — *Seattle, WA.*

- Spearheaded parallel experimentation for **Alexa+**, implementing traffic splitting and consistent hashing **(Kotlin/Guava)** to boost ML experimentation velocity by **96% (days to minutes).**
- Modernized **Alexa+** experimentation platform from single-lane to multi-tenant, boosting experimentation accuracy by **60%** by implementing deterministic identity routing and isolation guardrails.

AI Researcher — Sept. 2024 – May 2025
Stanford Artificial Intelligence Laboratory — *Remote*

- Expanded Bengali NLP evaluation resources by translating **20 major LLM benchmark datasets** into Bengali, enabling systematic evaluation of models for an underrepresented language spoken by over **250 million speakers.**
- Identified primary failure modes limiting Bengali LLM quality via **10-model, 8-benchmark evaluation**; tokenizer analysis surfaced materially higher tokens/word versus English, tracing issues to Unicode normalization and vocab coverage.

Software Engineering Intern — May 2024 – Sept. 2024
Amazon AGI — *Seattle, WA.*

- Drove platform modernization with a model release pipeline that enforced semantic versioning and automated checks, **compressing time-to-release by 60%** and streamlining the path from training to production.
- Operationalized automated testing end-to-end (pipeline regressions and post-deploy checks), **achieving a 90% reduction in update time** and solidifying model reliability across versions.

Machine Learning Engineer — Sept. 2023 – May 2024
Sage AI — *Remote*

- Amplified customer engagement by **42%** through **end-to-end ownership** of a full-stack CRM (**React**, **Rust**, **DynamoDB**, **Cognito**), delivering secure onboarding, real-time analytics and zero-downtime releases.
- Engineered a production-grade SEC filings ingestion pipeline in Python coupled with LLM-driven financial analysis, increasing **analysis speed/precision by 23%** and raising **bank decision accuracy by 36%**.

Research Assistant — Aug. 2023 – May 2024
RIT Computer Science — *Rochester, N.Y.*

- Increased student engagement by **25%** through an **NSF(National Science Foundation)-funded** rollout of an R sandbox and R interactive sessions alongside Python, enabling side-by-side comparison of data-science learning outcomes.
- Enhanced platform interactivity by **15%** by integrating **Thebe** and **Binder** with version-locked **R/Python** kernels for reliable in-browser execution and reproducible labs.

Research Assistant — Jan. 2022 – May 2022
RIT Computer Science - Dr. Sumita Mishra — *Rochester, N.Y.*

- Simulated realistic energy-grid communication using an improved **Shamir's Secret Sharing Scheme** on **Raspberry Pis**, documenting implementation gaps between theoretical guarantees and practical deployment.
- Created comprehensive implementation guidelines that helped subsequent researchers avoid common pitfalls, contributing to the reproducibility and accessibility of cryptographic research.

Software Engineering Intern — May 2023 – Aug. 2023
Amazon AGI — *Seattle, WA.*

- Led design and rollout of a rubric-driven feedback pipeline and evaluation dashboards for Alexa's next-gen Q&A LLM, **enhancing model performance by 9%** under standardized evaluation criteria.
- Decreased end-to-end model testing time by **12%** through a parallel, tabbed multi-LLM interface enabling concurrent prompts and rapid comparison as verified by Applied Scientists.

Projects

Vision Voice (Link) | *HuggingFace, BLIP, TTS, Gemini, FastAPI, Angular*

- Implemented end-to-end pipelines for image-to-text, text summarization and text-to-speech by integrating **BLIP** and **Gemini** on **HuggingFace** to generate accessible image descriptions.
- Deployed SpeechT5-based **TTS** to convert captions to audio, delivering **up to 7.5 hours/day** in potential time savings for audio-description workflows.

Math Mania (Link) | *Machine Learning, TensorFlow, TensorBoard, JavaScript, HTML/CSS*

- Built an interactive platform for my niece, turning drills into game-like challenges with progress tracking to increase engagement.
- Designed and trained from scratch a **TensorFlow model** on **MNIST** dataset for digit classification using **TensorBoard** to refine architecture and hyperparameters.

Teaching & Leadership

Section Leader Apr. 2023 – Jun. 2023
Stanford University - Code in Place *Remote*

- Mentored students in Python programming fundamentals, providing personalized feedback and guidance through weekly interactive sessions to strengthen problem-solving skills and programming confidence.

Student Lab Instructor Aug. 2022 – May 2023
Rochester Institute of Technology *Rochester, N.Y.*

- Guided students through hands-on research projects, fostering critical thinking and technical proficiency while maintaining an inclusive learning environment that encouraged collaboration and peer learning.

Resident Advisor Jan. 2022 – May 2024
Rochester Institute of Technology *Rochester, N.Y.*

- Led a residential community of **40+ students**, facilitating conflict resolution, organizing educational programming, and fostering an inclusive environment that supported academic success and personal development.

Technical Skills

Programming Languages: C, C++, Java, Python, JavaScript, TypeScript, SQL, Rust, C#, .NET, Go, PHP, Swift, Kotlin
Frameworks & Libraries: Spring Boot, Node.js, Express.js, React, Next.js, Fast API, Django, GraphQL, Jest, JUnit
Databases & Data Tools: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Firebase, Pandas, NumPy, Matplotlib
Cloud, DevOps & Testing: Docker, Kubernetes, AWS, GCP, Azure, CI/CD, Git, GitHub
Machine Learning & Other: RLHF, NLP, LLM, Generative AI, TensorFlow, PyTorch, Keras, Scikit-learn, Transformers, LangChain, OpenCV, OpenAI API, OAuth, Apache Spark

Honors, Awards & Publications

Publications:

• Bhowmik, S. et al. "Evaluating LLMs' Multilingual Capabilities for Bengali: Benchmark Creation and Performance Analysis"

Honors & Awards:

• Outstanding Undergraduate Scholar Award (Top 1% of undergraduates) - RIT, May 2025
• Alumni Endowed Scholarship Award - RIT Department of Computer Science, Apr. 2025
• Alfred L & Ruby C. Davis Leadership Award - RIT, Mar. 2025
• Excellence in Student Life Award - RIT, Mar. 2025
• International Student Service Award - RIT, Mar. 2025
• Alfred L & Ruby C. Davis Leadership Award - RIT, Apr. 2024
• International Student Service Award - RIT, Apr. 2024
• Outstanding 1st Year Student Scholarship Award - RIT, Aug. 2022
• Asia Pacific ICT Award (APICTA) - China, Oct. 2018
• Bangladesh Association of Software & Information Services - National ICT Award, Jul. 2018
• Dean's List - RIT (All Semesters)
• NSF-funded Research Assistant - National Science Foundation